Energy-Efficient Computing

Homework 3

In this homework you will write a few small test programs in Python, Java, and C++ and compare how much energy each one uses.

Setup. On a UTCS lab machine, clone the repo and work inside it:

```
git clone https://github.com/ATX24/hw_3_eec.git
cd hw_3_eec
```

To measure a program, put `./rapl` in front of the command you would normally use to run it:

```
./rapl python3 test.py
./rapl java Test
./rapl ./test
```

`rapl` prints how long your program ran and how much energy the CPU and memory used. Add `-r 5` to run it 5 times. If `./rapl` does not run, rebuild it with `make`.

Write each program the same way in all three languages. Do not use libraries that do the work for you (like NumPy or built-in sort functions). Compile C++ with `g++ -O2`.

1. Write a Python, Java, and C++ version of each program below. For each one, use `rapl` to get the energy of all three versions, then compare them.

   a. Matrix multiply: two 400×400 matrices using a triple loop.

   b. Quicksort: 2,000,000 random integers.

   c. Mergesort: 2,000,000 random integers.

   d. Fibonacci: `fib(35)` using plain recursion.

2. Why? Explain the differences you saw in Question 1. (Hint: run `./rapl sleep 10` to see how much energy the machine uses doing nothing.)

3. Research how garbage collection in the Java runtime affects energy usage.

4. Research RAPL (Running Average Power Limit) on Intel processors. How does the chip keep track of energy, and how can software read that data? Look at `rapl.c` to see how our tool does it.

Submit: your source files and a PDF with your results and answers.